Special products

1 Determine whether the following statements are true. If not, rewrite the right-hand side to make the statement true.

a $(8+6)^2 = 8^2 + 6^2$

b $(y-7)^2 = y^2 - 21y + 49$

2 Complete each of the following statements:

a $(x+3)^2 = x^2 - \square + 9$

b $(x-3)^2 = x^2 - \square + \square$

c $x^2 + 64 - \square = (x-8)^2$

d $(x+\square)^2 = x^2 + \square + 36$

e $(8x^{\square} - 9)^2 = 64x^4 - 144x^2 + 81$

f $(-9x+2)^2 = 81x^2 - \square + \square$

g $(2y+\square)(2y-\square) = 4y^2 - 4$

h $(3x-8)^2 = 9x^2 - \square + 64$

3

a Find the product $(t+5)^2$.

b Find the product $(-t-5)^2$.

c Determine whether the following are equal to $(-t-5)^2$.

i $t^2 - 10t + 25$

ii $(-t)^2 + 10t + (-5)^2$

iii $-(t+5)^2$

iv $(t+5)^2$

4 Find the product of each of the following:

a $(x+3)^2$

b $(m-5)^2$

c $(s+4t)^2$

d $(x^2+3)^2$

e $(9x+2)^2$

f $(4-5m)^2$

g $(4x+7y)^2$

h $3(3x+8y)^2$

i $9(3x+8y)^2$

j $3x(7x-5y)^2$

k $\left(\frac{8x-9}{9}\right)^2$

l $\left(6y+\frac{1}{2}\right)^2$

m $\left(\frac{n}{4} - \frac{4}{n}\right)^2$

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5 Find the product of each of the following:

a $(m+3)(m-3)$

b $(x-7)(x+7)$

c $(3x-8)(3x+8)$

d $(4x-3y)(4x+3y)$

e $(4y+9)(4y-9)$

f $(6-p)(6+p)$

g $(-7y-8)(-7y+8)$

h $(11-9p)(11+9p)$

i $(10-m)(m+10)$

j $6(8x-9y)(8x+9y)$

k $5x(2x-9y)(2x+9y)$

l $(0.8x+9)(0.8x-9)$

m $(9x-0.4)(9x+0.4)$

n $(0.2x-11)(0.2x+11)$

o $\left(v+\frac{1}{7}\right)\left(v-\frac{1}{7}\right)$

p $\left(6x+\frac{1}{3}\right)\left(6x-\frac{1}{3}\right)$

q $\left(6x-\frac{8}{3}\right)\left(6x+\frac{8}{3}\right)$

r $\left(x-\frac{2}{x}\right)\left(x+\frac{2}{x}\right)$

s $a^2+(11-a)(11+a)-10$

6 Evaluate 403^2 by expressing 403 as the sum of 400 and 3, and then evaluating the product. Show your work.

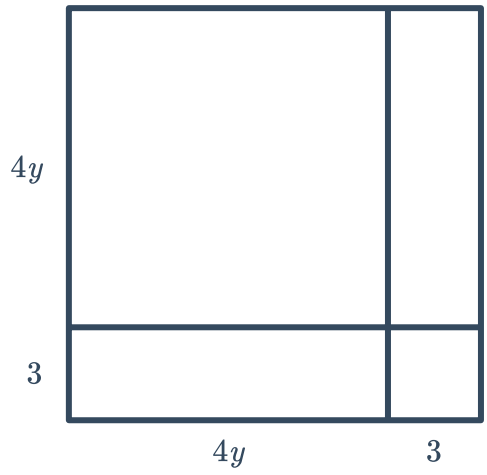
7 Evaluate 896^2 by expressing 896 as $900-4$, and then evaluating the product. Show your work.

8 Find the value of $201 \cdot 199$ by rewriting it in the form $(x+y)(x-y)$. Show your work.

9 Consider the following figure.

a Find the area of the large square by multiplying its side lengths.

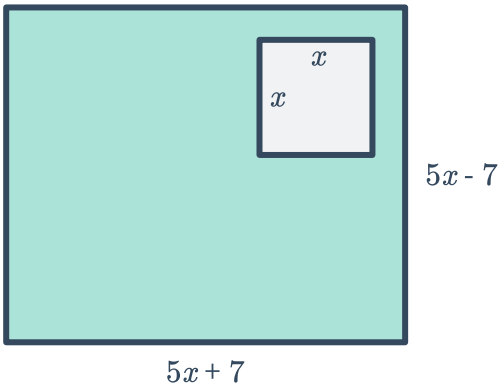
b Find the area of the large square by adding the areas of the smaller rectangles inside it.



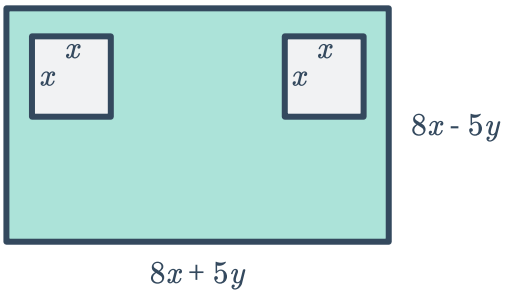
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10 For each of the following diagram, form an expression for the shaded area.

a



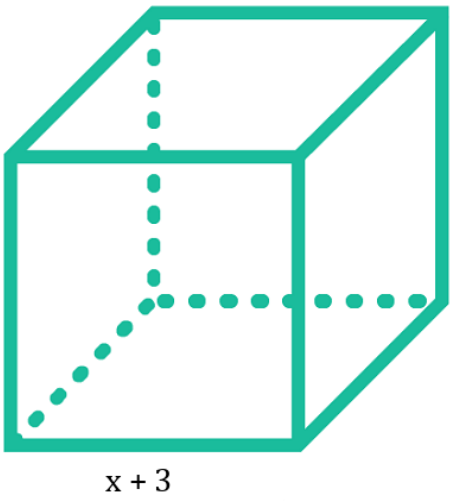
b



11 Consider the following cube.

a Find the surface area of the cube.

b Find the volume of the cube.



12 Write down an expression for the product of three consecutive integers if the middle integer is m.

13

a Find the product $\left(y^2+9\right)\left(y^2-9\right)$.

b Find an expression for the total distance traveled when traveling at $\frac{1}{y}\left(y^2+9\right)$ kilometers per hour for $y^2\left(y^2-9\right)$ hours.

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Additional questions

14 Prove the following results about special products:

a $(a+b)^2 = a^2 + 2ab + b^2$

b $(a-b)^2 = a^2 - 2ab + b^2$

c $(a+b)(a-b) = a^2 - b^2$

15 Show how you can use one of the special products $(a \pm b)^2$ to find each perfect square without a calculator.

a 21^2

b 19^2

c 33^2

d 47^2

16 A square with side lengths measuring $x - 1$ centimeters has each side enlarged by a factor of 2. Find an expression, for the area of the new square.

17 Show how the difference of squares pattern can be used to make the problem $105^2 - 104^2$ easier to evaluate without the use of a calculator.

18 Polynomial multiplication can be used to find the total area of a rectangle that has a border, where the length and width each represent a factor in the product.

a Write an algebraic model that can be used to find the total area of a square with an unknown side length and a border with an unknown border width.

b Use your model to find the total area of a square with a side length of $2m + 1$ and a border width of 3

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